

2024 HKDSE Physics

Paper 2 — Elective Sections

Original questions and official answers with detailed English reasoning. Calculations show relationships, substitution and units; each structured subpart is presented separately.

Editable questions with original diagrams, official answers and detailed English reasoning. Each subpart is separated for clarity.

This transcription includes all four elective sections. In the examination, candidates answer any two sections.

Section A: Astronomy and Space Science

Q.1: Multiple-choice questions

Official answers are placed directly below each question.

Question 1.1

The size of a typical galaxy is

- (1) larger than that of a typical star cluster.
- (2) smaller than that of a typical cluster of galaxies.
- (3) smaller than that of a typical nebula.

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

Official answer: A

A typical galaxy contains many star clusters and nebulae, while a cluster of galaxies contains many galaxies.

Thus a galaxy is larger than a typical star cluster but smaller than a typical galaxy cluster; it is not smaller than a typical nebula. Statements (1) and (2) are correct. Answer: A.

Question 1.2

If the Earth's mass were doubled and continued to move in its present orbit around the Sun, its orbital period

A. would increase.

B. would remain unchanged.

C. would decrease.

D. cannot be determined with the information given.

Official answer: B

Equating gravitational and centripetal forces gives $GM_{\text{sm}}/r^2 = 4\pi^2 mr/T^2$. The Earth's mass m cancels, so $T = 2\pi\sqrt{(r^3/GM_{\text{s}})}$. With the same orbit and solar mass, the period is unchanged even if Earth's mass doubles, within the usual fixed-Sun approximation. Answer: B.

Question 1.3

The escape velocity from Earth's surface is v . If an object is launched from the Earth's surface with initial speed $2v$ (assume the object's flight is unpowered except during launching), what will be its speed when the object is far away from the Earth? Neglect the effects of the Earth's atmosphere and of other celestial bodies.

A. $v/\sqrt{2}$ B. v C. $\sqrt{2} v$ D. $\sqrt{3} v$ **Official answer: D**

Escape speed satisfies $v^2 = 2GM/R$. Initially the specific total energy is $\frac{1}{2}(2v)^2 - GM/R = 2v^2 - \frac{1}{2}v^2 = \frac{3}{2}v^2$. Far away, gravitational potential energy tends to zero, so $\frac{1}{2}v_\infty^2 = \frac{3}{2}v^2$ and $v_\infty = \sqrt{3} v$. Answer: D.

Question 1.4

Which of the following descriptions about the Copernican model and Ptolemaic model of the solar system is/are correct?

- (1) The Copernican model is more accurate in predicting planetary motion.
- (2) The Copernican model's explanation of retrograde motion of planets is simpler.
- (3) The phases of Venus can be explained by both models.

A. (2) only

B. (3) only

C. (1) and (2) only

D. (1) and (3) only

Official answer: A

The original Copernican model retained circular motions and was not inherently more accurate than the Ptolemaic model. It explained retrograde motion more simply as the effect of relative orbital motion. The full observed range of Venus's phases is not explained by the classical Ptolemaic arrangement. Only statement (2) is correct. Answer: A.

Question 1.5

Figure (1) shows a binary system consisting of two stars orbiting in uniform circular motion about their centre of mass C . O is a distant observer on the same orbital plane of the stars. The time variation of the Doppler shift $\Delta\lambda$ of a particular spectral line from the two stars measured by O is shown in Figure (2). Time T on the graph is the instant corresponding to Figure (1). In what direction is star Y moving at time T ? Which curve, P or Q , is for the more massive star?

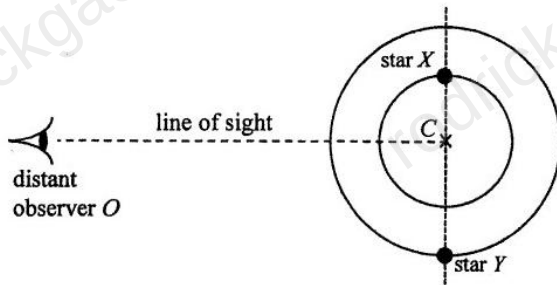


Figure (1)

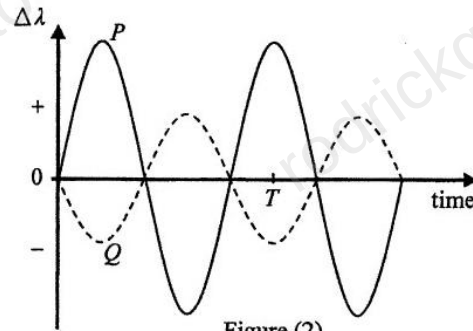


Figure (2)

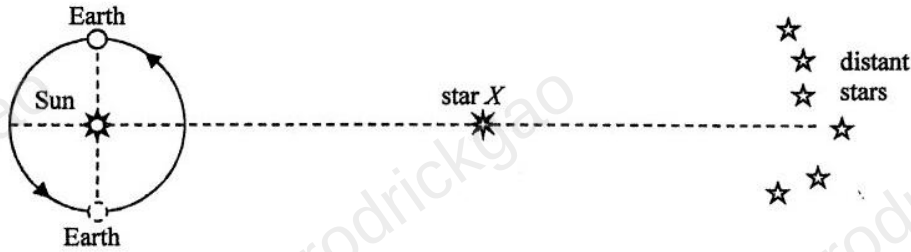
- A. towards O ; curve Q
- B. towards O ; curve P
- C. away from O ; curve Q
- D. away from O ; curve P

Official answer: C

Both stars share the same angular speed. Y has the larger orbital radius and therefore the larger speed and Doppler-shift amplitude, so it corresponds to curve P . At time T , P has positive $\Delta\lambda$, indicating recession from O . The more massive star lies closer to C , moves more slowly and corresponds to the smaller-amplitude curve Q .
Answer: C.

Question 1.6

If star X is 20 pc from the Earth, what is its maximum angular displacement against the distant stars when the star is viewed from the Earth 6 months apart?



- A. 0.025 arc seconds
- B. 0.05 arc seconds
- C. 0.1 arc seconds**
- D. 0.5 arc seconds

Official answer: C

The parallax angle is $p = 1/d = 1/20 = 0.05$ arc second. This is the half-angle based on a 1 AU baseline.

Observations six months apart use opposite ends of Earth's orbit, so the maximum angular displacement is $2p = 0.10$ arc second. Answer: C.

Question 1.7

A student has determined the apparent magnitude of star X to be m . After measuring the distance D to star X, the absolute magnitude of X is deduced to be M . However, after re-examining the data, D was in fact underestimated (i.e. D should be larger). What can be said about the actual apparent and absolute magnitudes of X?

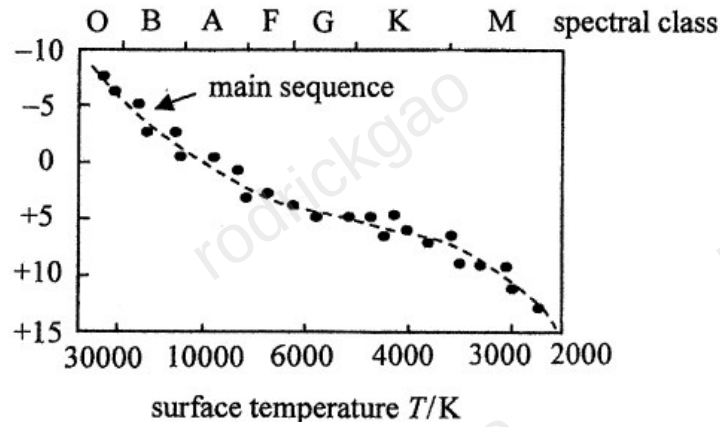
- A. apparent magnitude larger than m ; absolute magnitude larger than M
- B. apparent magnitude same as m ; absolute magnitude less than M**
- C. apparent magnitude less than m ; absolute magnitude less than M
- D. apparent magnitude same as m ; absolute magnitude larger than M

Official answer: B

The measured apparent brightness and apparent magnitude do not change when the distance estimate is corrected. From $m - M = 5 \log_{10}(d/10 \text{ pc})$, a larger d at fixed m implies a smaller M . The star is intrinsically more luminous than previously deduced. Answer: B.

Question 1.8

Star X is a main sequence star. It has a radius 8 times that of the Sun (a class G star). Its luminosity is 250000 times that of the Sun. Which spectral class does star X belong to?



- A. class M
- B. class K
- C. class G
- D. class O**

Official answer: D

Stefan's law gives $L \propto R^2 T^4$. Thus $(T_X/T_S)^4 = 250000/8^2 = 3906.25$ and $T_X/T_S \approx 7.91$. With the Sun near 5800 K, X is about 4.6×10^4 K, much hotter than class G and in spectral class O among the listed choices. Answer: D.

Q.1: Structured question

The Moon is the only natural satellite of the Earth and assume that it orbits the Earth in a circular orbit of radius $r = 384400 \text{ km}$.

Given: mass of the Earth $M_E = 5.97 \times 10^{24} \text{ kg}$; radius of the Moon $R_M = 0.273 \times \text{radius of the Earth} = 0.273 R_E$.

- (a) (i) Verify that the period of the Moon around the Earth is about 27 to 28 days. (3 marks)
- (ii) Given that the gravitational acceleration experienced by objects on the Earth's surface is about 6 times that on the Moon's surface, estimate the mass of the Moon M_M . (3 marks)
- (b) The amount of energy from the Sun reaching the Moon varies significantly as the position of the Moon relative to the Sun changes, leading to an average surface temperature at the Moon's equator highest at 390 K and lowest at 95 K. Suppose that I_H and I_L respectively are the highest and lowest intensity of radiation from the Sun on the Moon's surface. You may consider the Sun as the Moon's only energy source.
 - (i) Assuming that the Moon is a blackbody, estimate I_H/I_L . (2 marks)
 - (ii) State another assumption made for the above estimation besides assuming that the Moon is a blackbody. (1 mark)
- (c) Suppose a telescope is constructed on the Moon. State one advantage of such a telescope compared to that built on the Earth. (1 mark)

Official marking scheme

Solution	Marks	Remarks
1. (a) (i) Assuming a circular orbit, the orbital velocity $v = \frac{2\pi r}{T}$. As $\frac{GM_E M_M}{r^2} = \frac{M_M v^2}{r}$ $T^2 = \frac{4\pi^2 r^3}{GM_E} = \frac{4\pi^2 (384400 \times 10^3)^3}{(6.67 \times 10^{-11})(5.97 \times 10^{24})}$ $\Rightarrow T = 2.373039 \times 10^6 \text{ s}$ $= 27.465727 \text{ days } (\approx 27 \text{ to } 28 \text{ days})$ Or $\frac{GM_E}{r^2} = \frac{v^2}{r}$ $v = \sqrt{\frac{GM_E}{r}} = 1017.7905 \text{ m s}^{-1} \approx 1018 \text{ m s}^{-1}$ $T = \frac{2\pi r}{v} = \frac{2\pi(384400 \times 10^3)}{1018}$ $= 2.37 \times 10^6 \text{ s} \approx 27.5 \text{ days}$	1M 1M 1A 1M 1M 1A	Accept: $(2.33 \sim 2.38) \times 10^6 \text{ s}$
(ii) Gravitational acceleration on the Earth's surface $g_E = \frac{GM_E}{R_E^2}$ while that on the Moon is $g_M = \frac{GM_M}{R_M^2}$. Thus $\frac{g_E}{g_M} = \left(\frac{M_E}{M_M}\right) \left(\frac{R_M}{R_E}\right)^2 = \frac{5.97 \times 10^{24}}{M_M} \cdot (0.273)^2 = 6$ $\Rightarrow M_M = 7.415636 \times 10^{22} \text{ kg} \approx 7.42 \times 10^{22} \text{ kg}$	1M 1M 1A	Accept: $(7.4 \sim 7.42) \times 10^{22} \text{ kg}$
(b) (i) Let T_H and T_L be the highest and lowest average temperature on the Moon's surface, $L_H = \sigma(4\pi R_M^2)T_H^4$ $L_L = \sigma(4\pi R_M^2)T_L^4$ $\therefore \frac{I_H}{I_L} = \frac{L_H}{L_L} = \left(\frac{T_H}{T_L}\right)^4 = \left(\frac{390}{95}\right)^4 = 284.030 \approx 284$	1M 1A	
(ii) Assumption: Power / energy received / incident on the Moon is equal to power / energy radiated by the Moon.	1A	
(c) Accept any ONE: • No atmosphere to blur the images • No atmosphere to block / absorb various types of electromagnetic radiation • No atmosphere to scatter sunlight • No light pollution from artificial light • Observation can be made during day time or any time	1A	

(a)(i) Gravity provides the centripetal force: $GM_{\text{EMM}}/r^2 = M_M(4\pi^2 r/T^2)$. Therefore $T = 2\pi\sqrt{(r^3/GM_E)}$. With $r = 3.844 \times 10^8 \text{ m}$, $G = 6.67 \times 10^{-11}$ and $M_E = 5.97 \times 10^{24} \text{ kg}$, $T = 2.373 \times 10^6 \text{ s} = 27.47 \text{ days}$, consistent with 27–28 days.

(a)(ii) Since $g = GM/R^2$, $g_E/g_M = (M_E/M_M)(R_M/R_E)^2 = 6$. Thus $M_M = M_E(0.273)^2/6 = 7.42 \times 10^{22} \text{ kg}$. The Moon's radius must be squared in this ratio.

(b)(i) Assuming radiative equilibrium and blackbody emission, the incident power balances radiated power proportional to T^4 . Hence $I_H/I_L = (390/95)^4 = 284$. Kelvin temperatures must be used.

(b)(ii) The additional assumption is thermal/radiative equilibrium: at each temperature used, the power received from the Sun equals the power radiated by the Moon.

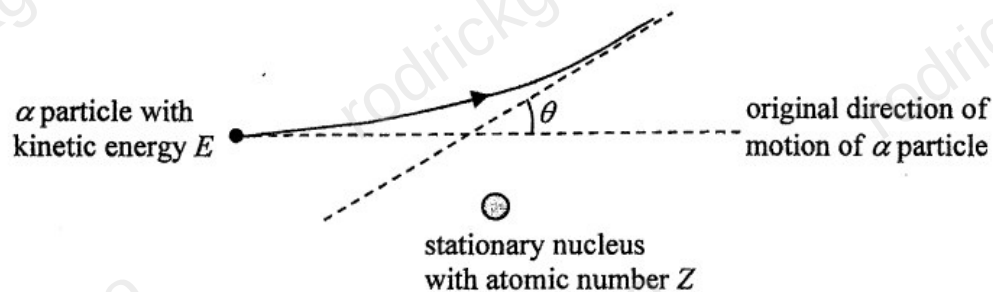
(c) With essentially no atmosphere on the Moon, atmospheric turbulence does not blur the telescope image. This gives a sharper image than for an otherwise comparable ground-based telescope on Earth.

Section B: Atomic World

Q.2: Multiple-choice questions

Question 2.1

Initially an α particle with kinetic energy E is far away from a stationary nucleus of atomic number Z . The α particle describes a curved trajectory making an angle of deviation θ with its original direction of motion as shown. Which of the following change in E and Z would give the greatest θ ?



- A. E increase; Z increase
- B. E increase; Z decrease
- C. E decrease; Z increase
- D. E decrease; Z decrease

Official answer: C

The positive nucleus repels the positive alpha particle. Increasing Z strengthens the electrostatic force; decreasing the incident kinetic energy makes the particle easier to deflect. At a fixed impact parameter, $\tan(\theta/2)$ is proportional to Z/E , so decreasing E and increasing Z gives the greatest deviation. Answer: C.

Question 2.2

When an electron collides with a gas atom and

- (1) if the collision is elastic, the electron transfers negligible kinetic energy to the atom.
- (2) if the atom is excited, all kinetic energy of the electron must be transferred to the atom.
- (3) if the atom is ionized, the atom must have absorbed the electron that collided with it.

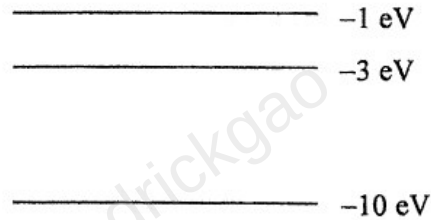
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: A

The atom is much heavier than the electron, so little kinetic energy is transferred to its translational motion in an elastic collision. Excitation requires only a particular energy-level difference; the electron can retain excess kinetic energy. Ionization removes a bound electron and does not require capture of the incident electron. Only (1) is correct. Answer: A.

Question 2.3

The figure shows the energy levels of an atom. Photons are emitted due to electron transitions between energy levels. Which of the following statements is/are correct? Given: the energy of photons in the visible spectrum lies between 1.78 eV and 3.10 eV.



- (1) Some of these photons are in the visible spectrum.
- (2) At most, photons of two different frequencies can be emitted.
- (3) The photons with the longest wavelength come from transitions between -1 eV and -10 eV .

A. (1) only

B. (2) only

C. (1) and (3) only

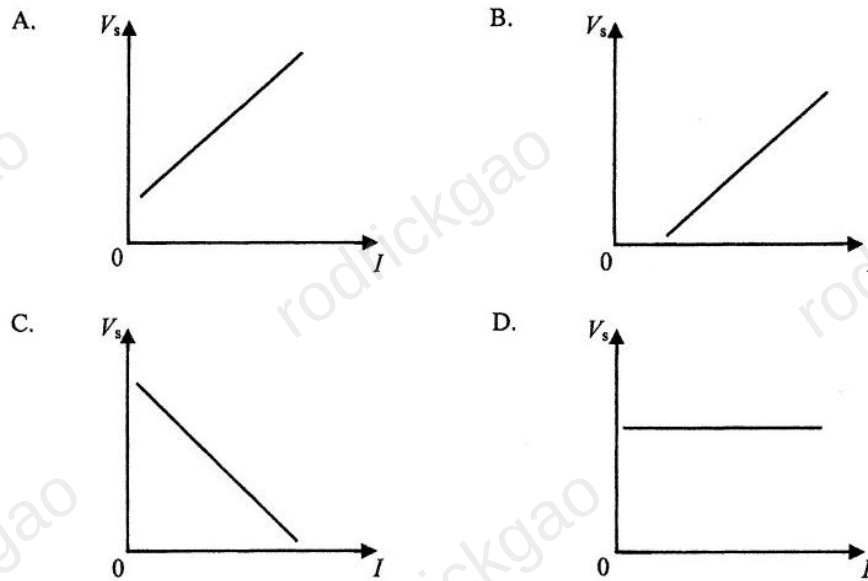
D. (2) and (3) only

Official answer: A

The three levels -1 , -3 and -10 eV give photon energies 2, 7 and 9 eV. Only 2 eV lies in the stated visible range, so (1) is true. There can be three distinct frequencies, not two. Since $\lambda = hc/E$, the longest wavelength corresponds to the smallest gap, -1 to -3 eV , not the 9 eV gap. Answer: A.

Question 2.4

In a photoelectric emission experiment, the variation of the stopping potential V_s of a metal with the intensity I of the monochromatic light used is studied. Which graph represents the experimental result?



- A. Graph A
- B. Graph B
- C. Graph C
- D. Graph D**

Official answer: D

Einstein's photoelectric equation is $eV_s = hf - \phi$. The light is monochromatic with fixed f and the metal has fixed work function ϕ , so stopping potential is independent of intensity. Increased intensity increases the number of emitted electrons, not their maximum kinetic energy. The horizontal nonzero graph is D. Answer: D.

Question 2.5

Metal X and Y are illuminated in turns by monochromatic lights of wavelengths λ_1 and λ_2 . Both metals show photoemission for λ_1 while only metal Y shows photoemission for λ_2 . Which deduction below is correct?

- A. $\lambda_1 > \lambda_2$ and the threshold frequency of X is higher.
- B. $\lambda_1 > \lambda_2$ and the threshold frequency of Y is higher.
- C. $\lambda_1 < \lambda_2$ and the threshold frequency of X is higher.**
- D. $\lambda_1 < \lambda_2$ and the threshold frequency of Y is higher.

Official answer: C

Only Y emits at λ_2 , so its work function and threshold frequency are lower than X's. Since λ_1 can also eject electrons from X, photons at λ_1 must have greater energy than those at λ_2 . With $E = hc/\lambda$, $\lambda_1 < \lambda_2$, and X has the higher threshold frequency. Answer: C.

Question 2.6

The de Broglie wavelength of an electron of mass m is equal to the wavelength of a photon that has energy E . Which of the following expressions is/are equal to the momentum of the electron? (c = speed of light in vacuum)

(1) $\sqrt{2mE}$

(2) mc

(3) E/c

A. (1) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

Official answer: B

Equal wavelengths imply equal momenta because $p = h/\lambda$ for both particles. The photon's energy is $E = hc/\lambda$, so the electron's momentum is E/c . E is the photon energy, not the electron's kinetic energy, so $\sqrt{2mE}$ is not justified; the electron's speed is not c . Only (3) is correct. Answer: B.

Question 2.7

In a transmission electron microscope (TEM), emitted electrons accelerate in an electron gun and then pass through components (1)–(3) as shown in the block diagram below before being projected through a lens and forming an image on a screen. Which of the following combinations is correct?



A. (1) condensing lens; (2) objective lens; (3) specimen

B. (1) condensing lens; (2) specimen; (3) objective lens

C. (1) objective lens; (2) specimen; (3) condensing lens

D. (1) specimen; (2) condensing lens; (3) objective lens

Official answer: B

The condenser/condensing lens concentrates the electron beam on the specimen. The transmitted electrons then enter the objective lens, which forms the first image before subsequent magnification and projection. The order is condensing lens $\rightarrow$ specimen $\rightarrow$ objective lens. Answer: B.

Question 2.8

Zinc oxide is a common ingredient in sunscreen products. Nanotechnology is often used in these products because

- A. zinc oxide in bulk is not able to block ultra-violet rays.
- B. zinc oxide in bulk is harmful to the skin.
- C. sunscreen products using nanosized zinc oxide particles are water repelling.
- D. sunscreen products using nanosized zinc oxide particles look transparent when applied on the skin.

Official answer: D

Nanosized zinc oxide particles are much smaller than visible wavelengths and scatter visible light much less strongly than bulk particles. The sunscreen can therefore look transparent while still blocking ultraviolet radiation. Bulk zinc oxide also blocks UV; water repellency is not the defining reason here. Answer: D.

Q.2: Structured question

Both Rutherford's atomic model and Bohr's atomic model picture the motion of an electron around the atomic nucleus.

- (a) Explain why Rutherford's atomic model is not stable. (2 marks)
- (b) (i) Name the kind of spectrum produced by a beam of white light passing through iodine vapour. (1 mark)
(ii) Referring to Bohr's atomic model, explain the formation of such kind of spectrum in (i). (2 marks)
- (c) Answer the following parts by referring to the energy level equation for hydrogen atom.
 - (i) Find the momentum of the emitted photon resulting from the transition of an electron from the 6th energy level (quantum number $n = 6$) of a hydrogen atom to the lowest energy level. (3 marks)
 - (ii) The energy E_n of the n th energy level of a hydrogen atom can also be expressed as $E_n = -K/r_n$, where r_n is the radius of the orbit at the n th energy level, K is a positive constant and E_∞ is taken to be zero. Express the electron's orbital radius r_6 at the 6th energy level in terms of the orbital radius r_1 at the lowest energy level. (2 marks)

Official marking scheme

Solution	Marks	Remarks
2. (a) In Rutherford's model, the electron emits electromagnetic waves and thus loses energy when it accelerates (orbits around the nucleus).	1A 1A	
	2	
(b) (i) absorption (line) spectrum	1A	
	1	
(ii) In Bohr's model, the electron can only orbit around the nucleus in certain allowed orbits and therefore having discrete energy levels. Only certain specific / discrete / quantized energies which equal the differences between these energy levels can be absorbed to produce the line spectrum.	1A 1A	
	2	
(c) (i) $\Delta E = -\frac{13.6 \text{ eV}}{6^2} - \left(-\frac{13.6 \text{ eV}}{1^2}\right)$ $= 13.222222 \text{ eV} = 13.222 \times (1.60 \times 10^{-19}) \text{ J}$ $\Delta E = \frac{hc}{\lambda_p} = pc$ Momentum of photon $p = \frac{\Delta E}{c}$ $= \frac{13.222 \times (1.60 \times 10^{-19})}{3.00 \times 10^8}$ $= 7.051852 \times 10^{-27} \approx 7.05 \times 10^{-27} \text{ kg m s}^{-1} \text{ or N s}$ Or $\Delta E = \frac{hc}{\lambda_p}$ $\lambda_p = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{13.222 \times (1.60 \times 10^{-19})}$ $= 9.401786 \times 10^{-8} \text{ m}$ $p = \frac{h}{\lambda_p} = \frac{6.63 \times 10^{-34}}{9.401786 \times 10^{-8}}$ $= 7.05 \times 10^{-27} \text{ kg m s}^{-1} \text{ or N s}$	1M 1M 1A	Accept: $(7.03 \sim 7.1) \times 10^{-27} \text{ kg m s}^{-1}$
	3	
(c) (ii) $E_1 = -\frac{13.6 \text{ eV}}{1^2} = -\frac{K}{r_1}$ $E_6 = -\frac{13.6 \text{ eV}}{6^2} = -\frac{K}{r_6}$ $\Rightarrow r_6 = 6^2(r_1) = 36 r_1$	1M 1A	
	2	

(a) In Rutherford's classical model, the orbiting electron is accelerating and should continuously emit electromagnetic radiation. It therefore loses energy and would spiral towards the nucleus rather than remain in a stable orbit.

(b)(i) An absorption line spectrum is produced.

(b)(ii) The allowed electronic energies are discrete. Photons can be absorbed only when hf equals an allowed energy-level difference, exciting an electron. These particular wavelengths are removed from the transmitted white light, producing dark absorption lines.

(c)(i) $E_n = -13.6/n^2 \text{ eV}$. The photon energy is $E_6 - E_1 = 13.6(1 - 1/36) = 13.2222 \text{ eV} = 2.11556 \times 10^{-18} \text{ J}$. For a photon $E = pc$, so $p = E/c = (2.11556 \times 10^{-18})/(3.00 \times 10^8) = 7.05 \times 10^{-27} \text{ kg m s}^{-1}$.

(c)(ii) $E_6/E_1 = 1/36$. Since $E_n = -K/r_n$, $E_6/E_1 = r_1/r_6$. Therefore $r_6 = 36r_1$.

Section C: Energy and Use of Energy

Q.3: Multiple-choice questions

Question 3.1

After using an electric hotplate and an induction cooker to cook the same dish, it is found that the surface temperature of the electric hotplate is much higher. The most probable reason is that



- A. the power of the electric hotplate is much higher.
- B. the electric hotplate heats up its plate and transfers thermal energy to the cooking utensil mainly by conduction.
- C. the process of electromagnetic induction does not produce thermal energy.
- D. the surface of the induction cooker is made of glass which is a poor conductor of heat.

Official answer: B

An electric hotplate first heats its own plate, which must be hotter than the utensil to transfer thermal energy to it mainly by conduction. An induction cooker produces heating directly in the metal utensil through induced currents; its surface is warmed secondarily by the utensil. The temperature comparison alone does not determine the appliances' power ratings. Answer: B.

Question 3.2

The energy labels of air-conditioners P and Q are shown below. The annual energy consumption (kW h) of each of them is based on 1200 hours of operation. Which one is (a) more energy efficient, (b) more suitable for a small room? (P: left; Q: right.)



- A. more energy efficient: Q; more suitable for a small room: P
- B. more energy efficient: P; more suitable for a small room: P
- C. more energy efficient: Q; more suitable for a small room: Q
- D. more energy efficient: P; more suitable for a small room: Q

Official answer: C

Use cooling output divided by electrical input, not the grade number alone. P has input power $2425/1200 = 2.02$ kW and cooling-to-input ratio $6.75/2.02 \approx 3.34$. Q has input $343/1200 = 0.286$ kW and ratio $2.18/0.286 \approx 7.63$, so Q is more efficient. Its smaller cooling capacity, 2.18 kW, also suits a small room better. Answer: C.

Question 3.3

Which component(s), evaporator, compressor, condenser and expansion valve, is/are located in the indoor unit of a split-type air-conditioner?

- A. expansion valve, evaporator and condenser
- B. expansion valve and compressor
- C. evaporator
- D. compressor

Official answer: C

The indoor evaporator absorbs thermal energy from room air as the refrigerant evaporates. The compressor and condenser belong to the outdoor unit in the split-system arrangement used in this syllabus, and the expansion valve is not an indoor component in the stated arrangement. The listed indoor component is the evaporator.

Answer: C.

Question 3.4

Which of the factors below determine the thermal transmittance (U-value) of a wall?

- (1) the thickness of the wall
- (2) the total area of the wall
- (3) the thermal conductivity of the wall's building material

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

Official answer: B

For a uniform wall, the thermal transmittance is $U = k/d$, where k is thermal conductivity and d is thickness, neglecting surface resistances. U is a property per unit area, so it does not depend on the total wall area.

Thickness and material conductivity determine it: (1) and (3). Answer: B.

Question 3.5

Compared to common glass windows, which of the following about a double-glazed window is/are correct?

- (1) Radiation penetrating the double-glazed window is significantly reduced.
- (2) The gas trapped in the double-glazed window has a much lower thermal transmittance than glass.
- (3) Double-glazed windows are not suitable for use in cold areas as more thermal energy would be lost through them.

A. (2) only

B. (3) only

C. (1) and (2) only

D. (1) and (3) only

Official answer: A

The trapped gas layer has low thermal conductivity and adds thermal resistance, reducing heat transfer through the window. Ordinary double glazing does not by itself imply a significant reduction of transmitted radiation; that depends on coatings and optical properties. It reduces heat loss in cold weather. Thus only statement (2) is correct. Answer: A.

Question 3.6

A container office of internal volume 37.5 m^3 is to be cooled from 33°C to 25°C by air-conditioners of total cooling capacity 8.80 kW . Estimate the time required if the container office has an exposed surface area of 57.5 m^2 and its Overall Thermal Transfer Value (OTTV) is 110 W m^{-2} . Given: density of air $= 1.20 \text{ kg m}^{-3}$; specific heat capacity of air $= 1000 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$.

- A. 24 s
- B. 41 s
- C. 118 s
- D. 145 s**

Official answer: D

The air mass is $\rho V = 1.20 \times 37.5 = 45.0 \text{ kg}$. Cooling requires $Q = mc\Delta T = 45.0 \times 1000 \times 8 = 3.60 \times 10^5 \text{ J}$. Heat enters at $\text{OTTV} \times \text{area} = 110 \times 57.5 = 6325 \text{ W}$, so net cooling power is $8800 - 6325 = 2475 \text{ W}$. Therefore $t = Q/P_{\text{net}} = 145 \text{ s}$. Ignoring the continuing heat gain would underestimate the time. Answer: D.

Question 3.7

On a particular day, steady wind of speed 8.0 m s^{-1} blows normally to a wind turbine for 12 hours while for the rest of the day the wind speed is too low for operation. The maximum energy generated that day is equivalent to a day with a steady wind of speed of _____ throughout the day.

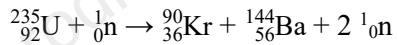
- A. 6.3 m s^{-1}**
- B. 5.0 m s^{-1}
- C. 4.0 m s^{-1}
- D. 3.2 m s^{-1}

Official answer: A

Available wind power is proportional to v^3 for fixed air density and swept area. Equate the daily energies: $8.0^3 \times 12 = v^3 \times 24$. Hence $v = 8.0 \times (1/2)^{1/3} = 6.35 \text{ m s}^{-1} \approx 6.3 \text{ m s}^{-1}$. The arithmetic mean speed is not appropriate because the power is cubic in speed. Answer: A.

Question 3.8

The equation below represents a typical fission reaction in a nuclear power plant.



Estimate the energy released when 1.00 kg of U-235 undergoes this fission reaction. The binding energies per nucleon of U-235, Kr-90 and Ba-144 are 7.59 MeV, 8.59 MeV and 8.26 MeV respectively. The mass of one mole of U-235 is 235 g.

A. 2.38×10^{25} MeV

B. 1.08×10^{26} MeV

C. 4.56×10^{26} MeV

D. 4.58×10^{26} MeV

Official answer: D

Energy released per fission is the increase in total binding energy: $\Delta E = 90 \times 8.59 + 144 \times 8.26 - 235 \times 7.59 = 178.69$ MeV. Free neutrons contribute no nuclear binding energy. The number of U-235 nuclei in 1.00 kg is $(1000/235)(6.02 \times 10^{23})$. Total release = 4.58×10^{26} MeV. Multiply binding energy per nucleon by the appropriate nucleon count before taking the difference. Answer: D.

Q.3: Structured question

Figure 3.1 shows a typical solar street lamp which consists of a solar panel, a storage battery and an LED lamp.

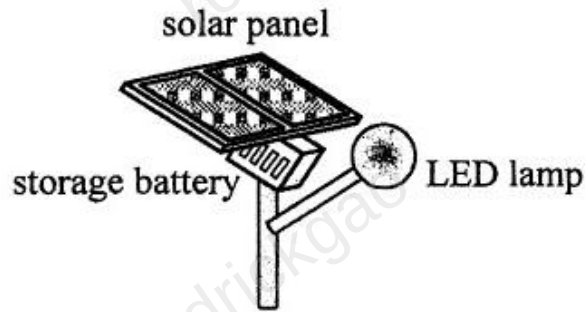


Figure 3.1

- (a) Describe the stages of conversion of solar energy involved when the storage battery is being charged. (1 mark)
- (b) The solar panel has an area of 0.111 m^2 and the battery can store a maximum of 405 W h energy. The overall efficiency of the charging system is 25%. Given: the maximum solar radiation power per unit area reaching the Earth's surface is 980 W m^{-2} .
- (i) The battery is completely discharged initially. Estimate the minimum charging time, in hours, for the solar panel to fully charge up the battery. (2 marks)
- (ii) The rated power of the LED lamp is 30 W . Suppose 80% of the energy stored in the battery can be used for lighting. Estimate the maximum duration of lighting in hours. (2 marks)
- (c) Such solar street lamps (rated power 30 W each) are installed at a height of 4.5 m above the ground along a road as shown in Figure 3.2.

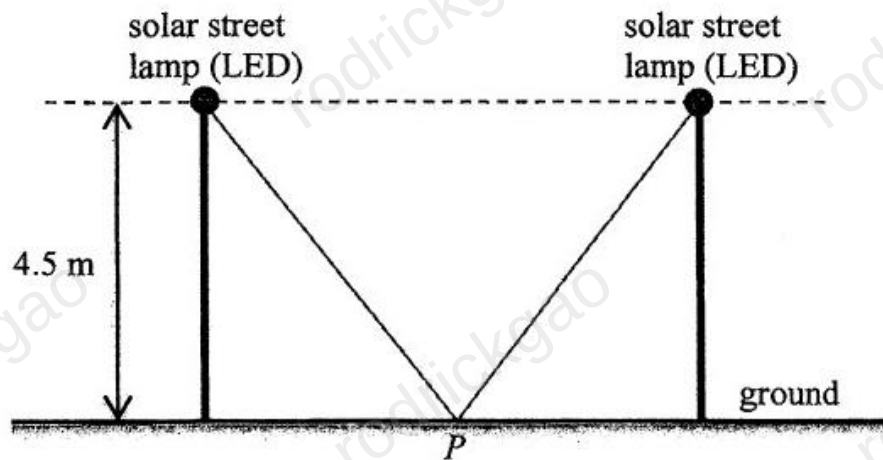


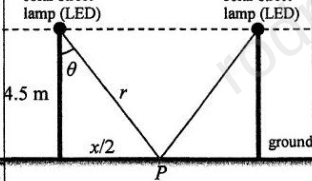
Figure 3.2

Each street lamp gives a luminous flux of 5160 lm .

- (i) Find the efficacy of such a street lamp in lm W^{-1} . (1 mark)
- (ii) The illuminance on the ground mid-way (P) between two adjacent street lamps should be at least 10 lx . Estimate the maximum separation between adjacent street lamps. (3 marks)

- (d) Although the efficacy of compact fluorescent lamps is comparable to that of LED lamps, LED lamps are preferred for systems employing solar panels. Explain. (1 mark)

Official marking scheme

Solution	Marks	Remarks
3. (a) (solar energy $\rightarrow$) electrical energy $\rightarrow$ chemical energy	1A	Accept: 14.9 ~ 15 (hours) Accept: (11.1 ~ 11.3) m 
(b) (i) $E = Pt$ $405 = (980 \times 0.111 \times 0.25) t$ $t = 14.892443 \approx 14.9$ (hours)	1M 1A	
(ii) $\frac{405 \times 0.8}{30}$ $= 10.8$ (hours)	1M 1A	
(c) (i) $\frac{5160}{30} = 172$ (lm W ⁻¹)	1M/1A	
(ii) $2 \times \frac{5160}{4\pi(4.5)^2} \times \cos^3 \theta = 10$ $\cos \theta = 0.627073$ $\theta = 51.165466^\circ$ $\tan \theta = \frac{x/2}{4.5}$ $x = 11.179937 \text{ m} \approx 11.2 \text{ m}$	1M 1M 1A	
Or $2 \times \frac{5160}{4\pi r^2} \times \frac{4.5}{r} = 10$ $r^3 = 369.558$ $r = 7.176193 \text{ m}$ $\frac{x}{2} = \sqrt{r^2 - 4.5^2}$ $x = 11.2 \text{ m}$	1M 1M 1A	
(d) The output voltage of the solar panel is d.c. LED lamps work on d.c. while (some) compact fluorescent lamps (CFL) work on a.c. / may need high operating voltage. Or LED has no toxic materials / CFL contains mercury (poses disposal issue) Or LED has a longer lifetime / replacement of LED is less frequent	1A	
	1	
	3	
	1	

(a) Solar radiation is converted to electrical energy by the solar panel, then to chemical energy stored in the battery.

(b)(i) The maximum storage power is $P = 980 \times 0.111 \times 0.25 = 27.195 \text{ W}$. Using $E = Pt$ with E in W h gives $t = 405/27.195 = 14.9 \text{ h}$. This is a minimum because it assumes maximum irradiance throughout charging.

(b)(ii) Usable lighting energy is $0.80 \times 405 = 324 \text{ W h}$. Therefore $t = 324/30 = 10.8 \text{ h}$.

(c)(i) Luminous efficacy = luminous flux/electrical power = $5160/30 = 172 \text{ lm W}^{-1}$.

(c)(ii) Let x be the lamp separation and r the distance from either lamp to midpoint P . Assuming isotropic point sources, each contributes $[5160/(4\pi r^2)] \cos \theta$, with $\cos \theta = 4.5/r$. At the limiting illuminance, $2 \times 5160 \times 4.5 / (4\pi r^3) = 10$, giving $r = 7.176 \text{ m}$. Thus $x = 2\sqrt{(r^2 - 4.5^2)} = 11.18 \text{ m} \approx 11.2 \text{ m}$. Include both lamps and the oblique-incidence factor.

(d) The solar panel and storage battery supply d.c. LED lamps operate from d.c. with a suitable driver, whereas conventional CFL systems may require a.c. or a higher operating voltage and additional conversion.

Section D: Medical Physics

Q.4: Multiple-choice questions

Question 4.1

The near point of Mary is at 68 cm. With her spectacle lenses held at 1.5 cm away from the eyes by the spectacle frame, she is able to read clearly texts 25 cm from her eyes. What is the power of her spectacle lenses?

- A. +2.25 D
- B. +2.75 D**
- C. +5.25 D
- D. +5.75 D

Official answer: B

Measure object and virtual-image distances from the spectacle lens: $u = 25 - 1.5 = 23.5$ cm and $v = -(68 - 1.5) = -66.5$ cm. The power is $P = 1/u + 1/v = 1/0.235 - 1/0.665 = +2.75$ D. The positive sign corresponds to a converging lens that forms a virtual image at Mary's near point. Answer: B.

Question 4.2

Which of the following statements about hearing is INCORRECT?

- A. The auditory canal of the outer ear only allows sound of audible frequencies to get through.**
- B. The middle ear is responsible for sound amplification.
- C. The inner ear is responsible for converting sound into electrical signals to the brain.
- D. Hearing involves hair cells being stimulated mechanically in the inner ear.

Official answer: A

The auditory canal conveys pressure waves to the eardrum; it is not a filter that admits only the audible frequency range. The middle ear provides mechanical amplification, while mechanically stimulated hair cells in the inner ear convert vibrations into signals for the brain. A is the incorrect statement. Answer: A.

Question 4.3

Which statement about a fibre optic endoscope is INCORRECT?

- A. Each optical fibre inside the endoscope is wrapped with material having a refractive index slightly lower than that of the fibre.
- B. The very small diameters of optical fibres ensure their flexibility for bending.
- C. The same bundle of optical fibres is used for illumination as well as image formation.**
- D. The resolution of image is limited by how thin the optical fibres are.

Official answer: C

A fibre-optic endoscope uses separate bundles for illumination and image transmission. The image bundle must preserve the relative positions of fibres. Lower-index cladding supports total internal reflection; thin fibres allow flexibility, and fibre size limits image resolution. The claim that one bundle performs both roles is incorrect. Answer: C.

Question 4.4

When new windows were put into a house, the sound intensity measured inside changed from I_1 to I_2 . Given that I_0 is the threshold sound intensity. The change in sound intensity level is given by

- A. $10 \log_{10}[(I_1 - I_2)/I_0]$
- B. $10 \log_{10}[(I_2 - I_1)/I_0]$
- C. $10 \log_{10}(I_2/I_0) - 10 \log_{10}(I_1/I_0)$
- D. $10 \log_{10}(I_1/I_0) - 10 \log_{10}(I_2/I_0)$

Official answer: C

Sound intensity level is $L = 10 \log_{10}(I/I_0)$. Its signed change is final minus initial: $\Delta L = 10 \log_{10}(I_2/I_0) - 10 \log_{10}(I_1/I_0) = 10 \log_{10}(I_2/I_1)$. If the new windows reduce the intensity, this change is negative. Answer: C.

Question 4.5

Tissue	Density / kg m^{-3}	Speed of sound / m s^{-1}
bone	1700	3780
fat	950	1450
muscle	1070	1590

The table lists the density of some tissues and the speed of sound in those tissues. Which statements about these tissues are correct?

- (1) Fat has the lowest acoustic impedance.
 - (2) Among the interfaces formed by the tissues listed, fat–muscle interface has the smallest intensity reflection coefficient.
 - (3) Bone–muscle interface and muscle–bone interface have the same intensity reflection coefficient.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

Official answer: D

Acoustic impedance $Z = \rho c$ gives bone 6.426×10^6 , fat 1.3775×10^6 and muscle $1.7013 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$, so fat has the lowest Z . For normal incidence, the intensity reflection coefficient is $[(Z_2 - Z_1)/(Z_2 + Z_1)]^2$. Fat and muscle have the smallest relative mismatch, and reversing the interface does not change the squared ratio. All three statements are correct. Answer: D.

Question 4.6

Linear attenuation coefficients (unit: cm^{-1}) of X-rays for different tissues are tabulated below.

Tissue	20 keV X-rays	50 keV X-rays	100 keV X-rays
bone	5.41	0.573	0.251
fat	0.517	0.193	0.154
muscle	0.872	0.240	0.178
lung	0.874	0.238	0.178

Determine which statement is INCORRECT according to the information given.

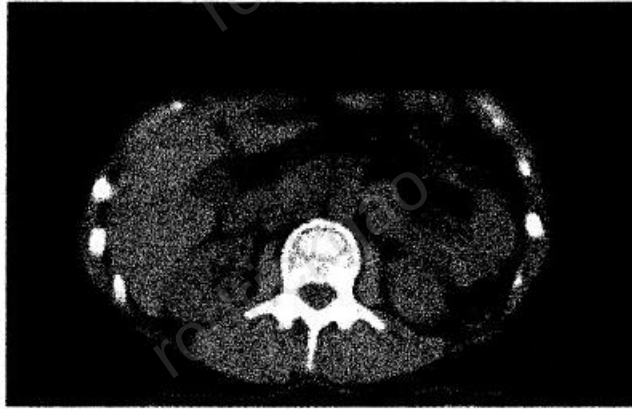
- A. Linear attenuation coefficient decreases with increasing energy of X-rays.
- B. Bone has the greatest decrease in X-rays absorption from low energy to high energy X-rays.
- C. 100 keV X-rays are the most suitable for X-ray radiography involving bone fractures.
- D. The intensity of 20 keV X-rays is approximately reduced to 13% of its initial value after passing through 10 cm thick muscle.

Official answer: D

For 20 keV X-rays in muscle, $\mu = 0.872 \text{ cm}^{-1}$. The transmitted fraction through 10 cm is $I/I_0 = e^{-\mu x} = e^{-0.872 \times 10} = 1.63 \times 10^{-4}$, or about 0.0163%, not 13%. Keep the thickness in centimetres to match μ . This directly contradicts statement D. Answer: D.

Question 4.7

Which of the following statements about the computed tomography (CT) image above is/are correct?



- (1) The image shows the absorption levels of X-rays by tissues.
- (2) The white region of the image indicates the highest level of reflection of X-rays.
- (3) The absorption levels of X-rays represent different functions of tissues.

- A. (1) only**
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: A

A CT image maps differences in X-ray attenuation/absorption and primarily shows anatomical structure. White regions have high attenuation, not high X-ray reflection. Attenuation values do not directly represent tissue functions as in functional radionuclide imaging. Only statement (1) is correct. Answer: A.

Question 4.8

Which of the following statements about radionuclide imaging (RNI) is/are correct?

- (1) Spatial resolution of RNI is better compared to ultrasound scan.
- (2) RNI uses radioisotopes that emit X-rays.
- (3) Most radioisotopes for RNI are artificially produced as they do not exist naturally.

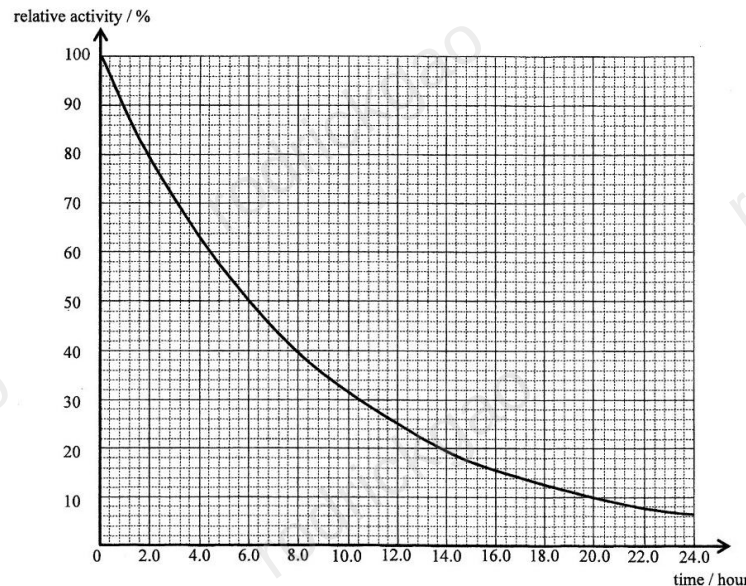
- A. (1) only
- B. (3) only**
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: B

In the standard RNI method considered here, gamma-emitting artificial tracers are detected outside the body. Its strength is showing tracer distribution and function, not superior spatial resolution to ultrasound. It does not use X-ray-emitting radioisotopes as stated in (2). Only statement (3) is correct. Answer: B.

Q.4: Structured question

- (a) α , β and γ rays pose radiation hazard to the human body. Referring to the penetrating power and ionizing power of radiation, explain why γ rays are chosen for radionuclide imaging (RNI). (2 marks)
- (b) The decay curve below represents the time variation of the relative activity of ^{99m}Tc (with background radiation already corrected for).



- (i) Find the 'physical half-life' of ^{99m}Tc (in hours). (1 mark)
- (ii) In an RNI study, a solution containing a ^{99m}Tc compound is prepared for injection to the patient concerned. The solution was calibrated at 9:00 am and found that each mL of solution has 550 MBq radioactivity. If the injection to the patient is scheduled at 11:00 am (i.e. two hours later) with a prescribed dosage of 600 MBq, find the volume of solution required, in mL. (3 marks)
- Given that the 'effective half-life' of ^{99m}Tc in a human body is 4.8 hours.
- (iii) Find the percentage of dosage left in the body 24 hours after the injection. (2 marks)
- (iv) According to the decay curve of ^{99m}Tc , the relative activity of ^{99m}Tc would decrease to about 6% in 24 hours. Account for the difference between the result found in (iii) and this value (i.e. about 6%). (2 marks)

Official marking scheme

Solution	Marks	Remarks
4. (a) <i>Penetrating power and its implication:</i> γ rays have no (rest) mass and are pure energy with highest penetrating power / penetrate longest (largest) distance in body than α and β particles; pass through body / can be detected using Gamma cameras outside the body.	1A	
<i>Ionizing power and its implication:</i> γ rays have least / smallest ionizing power, thus can be used safely as they mostly just pass through the body and cause less ionization damage / less harmful in our cells.	1A	
	2	
(b) (i) The physical half-life $T_{\frac{1}{2}}$ of ^{99m}Tc is 6 (hours).	1A	
	1	
(ii) Time interval t is 2 hours and A_0 is 550 MBq/mL, thus the activity A at 11:00 a.m. is		e.c.f. from (b)(i)
Alternatively: $600 = (550 \times V) \left(\frac{1}{2}\right)^{\frac{2}{6}}$ $V = 1.37 \text{ (mL)}$		Or $A = A_0 e^{-t \left(\frac{\ln 2}{T_{\frac{1}{2}}}\right)} = A_0 e^{-t(0.1155)}$ $600 = A_0 e^{-\frac{2 \ln 2}{6}} = A_0 \times 0.7937$ $\therefore A_0 = 755.953 \left(\frac{\text{MBq}}{\text{mL}}\right)$
$A = A_0 e^{-t \left(\frac{\ln 2}{T_{\frac{1}{2}}}\right)}$ $= A_0 e^{-\frac{2 \ln 2}{6}} = 550 \times 0.7937$ $\therefore A = 436.5 \left(\frac{\text{MBq}}{\text{mL}}\right)$	1M	The volume required $= \frac{A_0}{550} = \frac{756}{550} = 1.37446 \approx 1.37 \text{ (mL)}$
The volume required $= \frac{600}{436.5}$ $= 1.37446 \approx 1.37 \text{ (mL)}$	1M	
	1A	
Or From the graph: $A = 550 \times 0.78$ to 550×0.80 $= 429$ to $440 \left(\frac{\text{MBq}}{\text{mL}}\right)$	1M	Or Relative activity: 0.78 to 0.80 $A_0 = \frac{600}{0.80}$ to $\frac{600}{0.78}$ $= 750$ to $769 \left(\frac{\text{MBq}}{\text{mL}}\right)$
The volume required $= \frac{600}{440}$ to $\frac{600}{429}$ $= 1.36$ to 1.40 (mL)	1M	The volume required $= 1.36$ to 1.40 (mL)
	1A	
	3	
(iii) The dosage left after 24 hours is		$\left(\frac{1}{2}\right)^{\frac{24}{4.8}} = 0.03125 = 3.125\%$
$A = 600 \times e^{-24 \times \frac{\ln 2}{4.8}} = 600 \times e^{-2.4 \times 0.1444}$ $= 600 \times 0.0313 = 18.75 \text{ (MBq)}$ i.e. $\frac{18.75}{600} \times 100\% = 3.125\%$	1M	
	1A	
	2	
(iv) Biological elimination / metabolism / diffusion / excretion of radioactive substances (drugs) in body accounts for the difference.	1A	
Or The effective decay of radioactive substances in body depends on biological decay and physical decay.		
Thus, the actual decay / disappearance / drop / decrease of activity (percentage dosage) in body is more than the relative activity curve (physical decay) alone.	1A	
	2	

(a) Gamma rays have sufficient penetrating power to escape the body and reach an external gamma camera. Compared with alpha and beta radiation in the syllabus model, their weaker ionizing action reduces localized damage, although the dose must still be controlled. Both detectability and reduced tissue damage are required in the explanation.

(b)(i) The relative activity falls to one-half in 6 h, so the physical half-life is 6 h.

(b)(ii) Before injection, use the physical half-life only. At 11:00 am, activity per mL = $550 \times 2^{-2/6} = 436.5 \text{ MBq mL}^{-1}$. Required volume = $600/436.5 = 1.374 \text{ mL} \approx 1.37 \text{ mL}$.

(b)(iii) In the body, use the effective half-life 4.8 h. After 24 h, five effective half-lives have passed; the fraction remaining is $(1/2)^5 = 0.03125$, or 3.125% (activity 18.75 MBq).

(b)(iv) The body also removes the tracer by biological processes such as excretion. Physical decay and biological elimination act together, so activity in the body decreases faster than the physical-decay curve alone, which gives $(1/2)^4 = 6.25\%$ after 24 h.